✅ 1. Competency #6: Identify and describe the mathematical aspects of a real-world problem situation, including identifying significant variables

- Highest relevance.
- Students must extract from text and table:
  - Key variables: daily production (2,000; 6,000), fault rates (5%; 3%)
  - Derived quantities: number of faulty units = production × fault rate
- For example, to evaluate statement 1, they compute:
  - Faulty video players = 2,000 × 5% = 100
  - Faulty audio players = 6,000 × 3% = 180
- This process exemplifies recognizing core mathematical objects (quantities, ratios) and their relationships—the essential first step in modeling.

✅ 2. Competency #7: Simplify or decompose a situation or problem to make it amenable to mathematical analysis

- Highly relevant.
- The real-world context involves production, quality control, and repair—a complex system—but the problem has already been simplified to two product lines with given outputs and defect rates.
- Students must further decompose each statement into computable subtasks:
  - Statement 1 → compare two fault counts
  - Statement 2 → sum faults (100 + 180 = 280 ≠ 260)
  - Statement 3 → compute overall fault rate = 280 / (2,000 + 6,000) = 3.5%
- This translation of qualitative claims into basic arithmetic operations is precisely problem decomposition and simplification.

✅ 3. Competency #5: Recognize mathematical structure (including regularities, relationships, and patterns) in problems or situations

- Highly relevant.
- Students must identify fundamental mathematical relationships:
  - "Part = Whole × Percentage"
  - "Overall rate = Total faulty / Total produced"
- These reflect core structures of proportions and weighted averages.
- For instance, statement 3 tests whether students understand that the overall fault rate is not the simple average (5% + 3%) / 2 = 4%, but a weighted average due to unequal production volumes.
- Recognizing this structure determines correctness.

✅ 4. Competencies #10 / #11: Use mathematical tools (variables, symbols, diagrams) to describe mathematical structures and relationships

- Moderately relevant (though not explicitly required).
- While the response format only asks for "Yes/No," effective solvers will mentally or on scratch paper formulate expressions, such as:
  - $F_v = 2000 \times 0.05$
  - $F_a = 6000 \times 0.03$
  - $\text{Overall fault rate} = \frac{F_v + F_a}{8000}$
- This demonstrates using symbolic notation to represent real-world relationships, even if not written in the final answer.

🟡 5. Competency #12: Identify constraints, assumptions, and simplifications in a mathematical model

- Weak but present relevance.
- The problem relies on implicit assumptions:
  - Daily production and fault rates are stable "averages" (ignoring variability)
  - Fault occurrences are independent
  - Percentages can be directly applied to expected values
- Although students aren't asked to critique these, recognizing that "this model assumes average behavior" touches on this competency.

✅ 1. 能力 #6：识别并描述现实问题情境中的数学要素，包括识别关键变量

> *"Identify and describe the mathematical aspects of a real-world problem situation including identifying the significant variables"*

- 最高相关性。
- 学生必须从文字和表格中提取：
  - 关键变量：每日产量（2000, 6000）、故障率（5%, 3%）
  - 派生量：故障数量 = 产量 × 故障率
- 例如，判断“更多视频播放器故障”是否成立，需计算：
  - 视频故障数 = 2000 × 5% = 100
  - 音频故障数 = 6000 × 3% = 180
- 这一过程正是识别现实情境中的核心数学对象（数量、比率）及其关系，是建模的第一步。

✅ 2. 能力 #7：简化或分解情境，使其便于数学分析

> *"Simplify or decompose a situation or problem in order to make it amenable to mathematical analysis"*

- 高度相关。
- 原始情境涉及生产、质检、维修等复杂流程，但题目已将其简化为两个独立产品线的产量与故障率。
- 学生需进一步分解问题：将每个陈述拆解为可计算的子任务：
  - 陈述1 → 比较两个故障数量
  - 陈述2 → 计算总故障数（100 + 180 = 280 ≠ 260）
  - 陈述3 → 计算整体故障率 = 280 / (2000+6000) = 3.5%
- 这种“将复合陈述转化为基本算术操作”的过程，正是问题分解与简化的体现。

✅ 3. 能力 #5：识别问题或情境中的数学结构（如规律、关系、模式）

> *"Recognise mathematical structure (including regularities, relationships, and patterns) in problems or situations"*

- 高度相关。
- 学生需识别核心数学关系：
  - “部分 = 整体 × 百分比”
  - “总体比例 = 总故障数 / 总产量”
- 这些是比例与加权平均的基本结构。
- 例如，陈述3考查学生是否理解：整体故障率不是(5% + 3%) / 2 = 4%，而是加权平均（因产量不同）。
- 能否识别这一结构，直接决定答案正误。

✅ 4. 能力 #10 / #11：使用数学工具（变量、符号、图表）描述问题中的数学结构与关系

> *"Use mathematical tools (using appropriate variables, symbols, diagrams) to describe the mathematical structures and/or relationships in a problem"*

- 中等相关（尽管题目未要求书写公式，但隐含使用）。
- 虽然答题只需圈“Yes/No”，但有效解题者会在脑中或草稿上建立数学表达，例如：
  - $F_v = 2000 \times 0.05$
  - $F_a = 6000 \times 0.03$
  - $\text{Total fault rate} = \frac{F_v + F_a}{8000}$
- 这体现了用符号和运算表达现实关系的能力，即使未显性写出。

🟡 5. 能力 #12：识别数学模型中的约束、假设与简化

> *"Identify the constraints, assumptions simplifications in a mathematical model"*

- 弱相关但存在。
- 题目隐含假设：
  - 每日产量和故障率“平均”稳定（忽略波动）
  - 故障事件相互独立
  - 百分比可直接用于期望值计算
- 虽未要求学生反思这些假设，但若能意识到“这是基于平均值的理想化模型”，则触及此项能力。